

The Cyclical Night: Glitches, Governance, and the Erasure of Night in Google Street View

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Figure 1: Overview of the project La Noche Cíclica, showing all four components: the CRT video sculpture, interactive web archive, layered print album, and conceptual feedback intervention.

Abstract

“La Noche Cíclica” is a multi-part project that investigates the algorithmic suppression of night in Google Street View. Through a video sculpture, web archive, print album, and conceptual intervention, the work reveals how fragments of nighttime—caused by system delays and stitching errors—disrupt the platform’s default spatial-temporal logic of continuous daylight. These fleeting anomalies challenge the linear, functional temporality of machine vision and question the infrastructure behind digital mapping systems.

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The project explores how data-driven media reconstruct the visible world, and how moments of error can become sites of resistance, visibility, and alternative spatial memory.

CCS Concepts

• **Applied computing** → **Media arts**; *Sociology*; • **Social and professional topics** → **Corporate surveillance**; • **Information systems** → *Digital libraries and archives*.

Keywords

Media Infrastructure, Algorithmic Governance, Machine Vision, Capitalist Temporality

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1 Introduction

In an era where machine vision governs how we perceive and record the world, Google Street View (GSV) has become one of the most pervasive visual infrastructures shaping our collective understanding of space and time. More than a navigation tool, GSV functions as a curated and algorithmically optimized system that constructs a standardized visual reality. Through automated image stitching, AI-driven updates, and content moderation, GSV overwhelmingly privileges daytime imagery. Nighttime, with its low visibility and perceived unpredictability, is often excluded—filtered out as visual noise.

This default preference for daylight is not simply a technical decision, but an ideological one. GSV reflects what can be termed a “daytime bias” in machine vision, privileging brightness, legibility, and productivity. Night, by contrast, is rendered marginal or even invisible. Within capitalist temporal structures, night is treated not as its own time but as a transition between days—a seam, not a scene. Lighting technologies and surveillance systems increasingly colonize nighttime, attempting to prolong the utility of day and reinforce a temporal logic that aligns visibility with value. GSV’s near-complete omission of night imagery suggests a broader effort to erase temporal heterogeneity in favor of a continuous, productive, and controlled visual world.

And yet, the system is not seamless. Due to processing delays, data gaps, or algorithmic failures, GSV occasionally captures fleeting moments of night— anomalies that appear as visual glitches in the platform’s otherwise consistent daytime logic. These rare fragments—often brief, disjointed, and surrounded by overexposed or overwritten scenes—represent ruptures in the algorithmic logic of GSV. While easy to overlook, these nighttime appearances reveal the invisible labor of algorithmic governance and the underlying infrastructures that determine what is shown, when, and why.

This paper investigates these nighttime anomalies through an artistic research project titled *La Noche Cíclica*. The project is a multi-part installation consisting of a CRT-based video sculpture, an interactive web archive of nighttime coordinates, a print-based screenshot collection, and a conceptual intervention reporting these coordinates back to Google as “errors.” By tracing, capturing, and recontextualizing these visual fragments, *La Noche Cíclica* explores how digital infrastructures erase temporal difference, and how moments of failure or noise can become sites of aesthetic and political significance.

While prior research on GSV has examined its roles in urban visualization, privacy concerns, and mapping accuracy, few studies have addressed how it shapes our experience of temporality—particularly through the omission of night. This project positions itself at the intersection of media infrastructure, algorithmic vision, and artistic intervention. By foregrounding anomalies rather than seamless-ness, it proposes a methodological shift: to read visual failures as opportunities for critique, and to treat digital imperfections as sites of resistance.

1.1 Video Sculpture

The video sculpture of *La Noche Cíclica* is displayed through a 1970s Panasonic monochrome CRT television. Using the television’s channel dial, viewers switch between a globe in diurnal rotation and

a collection of GSV night coordinates, each capturing a different kind of algorithmic disruption. At the same time, they attempt to locate the embedded walkie-talkie reading of Borges’ *La Noche Cíclica* amid static and fragments of public broadcast. A vertical black line—the trace of CRT burn-in—appears faintly on the screen, signaling the collapse of night. These sculptural and interactive elements emphasize the tension between global systems of surveillance and fleeting visual failures, reclaiming night as a fractured but persistent presence.



Figure 2: Viewer interaction, switching channels to reveal different effects, video contents, and local radio broadcast.



Figure 3: Deliberate vertical burn-in line on the CRT display.

1.2 Web Archive

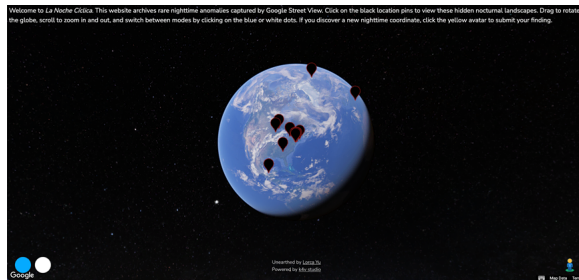
The project also includes an interactive online archive that indexes the geographic coordinates of nighttime scenes discovered within GSV. Each entry is manually located, captured, and annotated with metadata including location, date of last update, and name of founders. Visitors enter the archive through a dynamic portal that leads to specific night coordinates. Once inside, they can walk step by step through the glitch-embedded landscapes—immersing themselves in scenes where night briefly interrupts the platform’s continuous daylight. In contrast to GSV’s frictionless exploration, this navigational experience foregrounds the act of passage, inviting viewers to encounter temporal rupture not as error, but as a navigable terrain. The archive serves both as a poetic interface and as a data-driven tool for resisting infrastructural smoothness.



(a) 40° 39' 58" N, 73° 29' 12" W



(b) 18° 27' 51" N, 92° 44' 51" W

Figure 4: Screenshots of GSV nighttime coordinate.**Figure 5: Screenshot of the web archive portal. Viewers enter through this interface to access and navigate the collection of nighttime coordinates.****Figure 6: Viewer browsing the web-based archive, walking through nighttime coordinates in GSV. Their screen is projected onto the wall, transforming the act of navigation into a performative gesture.**

1.3 Print Album

A physical print album complements the digital archive by translating screen-based anomalies into a tactile, layered experience. Each page consists of two physically overlapping cards: a transparent film printed with elements from GSV interface—navigation arrows, toolbars, timestamps—and a separate layer of corresponding nighttime screenshot. When closed, the archive appears uniform, with each page containing a transparent card sleeve. Inside each sleeve are two layers: a transparent card printed with GSV interface elements and a corresponding nighttime screenshot beneath. At first glance, the layered composition resembles a typical GSV interface. But as the pages are turned, the transparent card can be removed from the sleeve, lifting away from the image beneath and visually as well as conceptually separating interface from content. This dual-layer structure foregrounds the constructed nature of machine-mediated vision. The transparency acts as a framing device, revealing the interface not as neutral, but as an active participant in shaping perception—one that can be peeled away and questioned. The nocturnal scenes underneath, isolated from their algorithmic wrapper, gain aesthetic presence and political weight.

1.4 Conceptual Intervention

The conceptual intervention engages directly with Google's data correction infrastructure. Each located nighttime coordinate is manually submitted to Google through its user feedback form, which invites viewers to suggest updates or corrections. This action transforms observation into disruption: it tests whether the system recognizes these images as "errors," and whether they are later overwritten by daytime imagery. In tracking the before-and-after status of reported locations, this conceptual gesture examines how algorithmic governance operates under the guise of visual maintenance. What constitutes a mistake? When does correction become erasure? The gesture reveals that even passive viewers on GSV are subtly conscripted into reinforcing normative visual standards—one feedback click at a time.

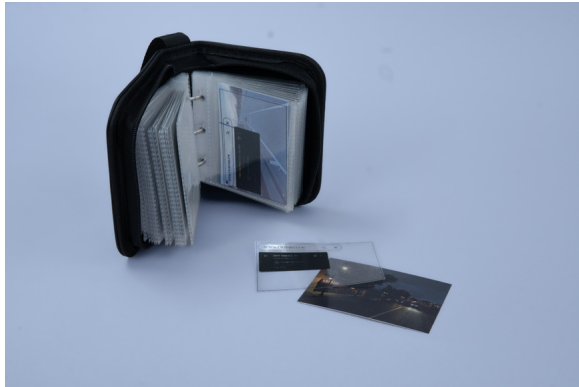


Figure 7: Overlay of transparent interface film and printed nighttime screenshot within the album.



Figure 8: Simulated “Report a Problem” interface embedded in the album, mirroring Google’s correction system.

2 Theoretical Framework

This study draws upon the media infrastructure framework developed by Lisa Parks and Nicole Starosielski, who define media not only as channels of content transmission but as deeply embedded socio-technical systems. Infrastructure often operates invisibly, only revealing its power structures and governing strategy when disruptions occur. The appearance of night in GSV is an example of this disruption on a platform designed to suppress such irregularities.

GSV exemplifies a visually oriented media infrastructure: it does not passively document reality, but actively constructs it through algorithmic stitching, data optimization, and spatial-temporal selection. This construction is not neutral; it reflects aesthetic preferences, technical constraints, and ideological assumptions embedded in platform design.

To understand the mechanisms and implications of GSV’s visual infrastructure, this section outlines three intersecting conceptual concerns:

2.1 Constructing Perception Through Infrastructure

GSV functions as a perceptual infrastructure, defining what counts as legitimate visibility. It filters reality through algorithmic processes that prioritize consistency and legibility. Machine learning and data correction routines ensure that only certain spatial and temporal conditions are retained, effectively constructing a curated and optimized world. By selectively presenting some times and discarding others, GSV normalizes a particular visual logic—one in which daytime becomes the standard, and all deviations are quietly erased. In this sense, GSV is not an archive but a visual regime.

2.2 Capitalist Temporality and the Erasure of Night

The erasure of night in GSV reflects broader capitalist temporal structures, in which productivity is tied to daylight and night is rendered residual. The project frames this erasure as a political and aesthetic act of exclusion, which is not merely the outcome of technical limitations, but a product of infrastructural governance. Like urban lighting systems or 24/7 surveillance networks, GSV participates in the colonization of night: a time stripped of its autonomy and repurposed to support continuous functionality. Within this context, night becomes a non-time, a gap to be filled, rather than a temporal condition.

2.3 Algorithmic Governance and the Classification of Error

GSV is a dynamic infrastructure, continually optimized through algorithmic updates. This project’s intervention—reporting nighttime coordinates as “errors”—tests how the system defines normalcy. The response of replacing night imagery with daylight enacts a form of algorithmic correction, revealing the governed visibility. This process raises critical questions: Who decides what qualifies as “error”? When do glitches become threats to infrastructural coherence? The act of correction becomes a site of governance, exposing how machine vision enforces normative visual regimes under the guise of optimization.

3 Literature Review and Related Work

This project engages with a broad body of scholarship across media studies, critical infrastructure theory, and artistic practice.

Foundational to the study is *Signal Traffic* by Lisa Parks and Nicole Starosielski, which conceptualizes infrastructure as a material, political force shaping everyday perception. Shannon Mattern’s *Deep Mapping the Media City* frames maps and networked visual systems as cultural and spatial technologies. Paul Virilio’s *The Vision Machine* critiques the political implications of automated seeing, while Kate Crawford’s *Atlas of AI* and Trevor Paglen’s *Invisible Images* expose the labor, bias, and opacity behind machine learning and algorithmic vision systems.

The temporal politics of visibility are further addressed in Jonathan Crary’s *24/7: Late Capitalism and the Ends of Sleep*, which traces how capitalism erodes night as a distinct temporal category. John Durham Peters’ *The Marvelous Clouds* offers a philosophical account of media as elemental infrastructure, while Henri Lefebvre’s *The*

Production of Space grounds spatial logic in socio-political context. Jenny Odell's *How to Do Nothing* contributes strategies for resisting infrastructural time regimes and reclaiming attention and locality.

Artistically, *La Noche Cíclica* draws influence from Rosa Menkman's glitch aesthetics, which celebrate visual anomalies as moments of resistance. Aram Bartholl's *How to Build a Fake Google Street View Car* subverts the aesthetics of surveillance infrastructure through mimetic critique. Karolina Sobiecka's *Last Clouds* and Jenny Odell's *Pipe Dreams* offer poetic meditations on the disappearance of natural or infrastructural phenomena in algorithmically rendered worlds.

This project extends these conversations by focusing on night in GSV—a temporality rarely addressed in existing discourse. Through aesthetic recontextualization and performative intervention, *La Noche Cíclica* contributes a methodological and conceptual approach to read errors not as failures, but as portals into the otherwise invisible operations of algorithmic visibility.

4 Technical Implementation

4.1 System Overview

The video sculpture features a 1979 model Panasonic TR-545 monochrome television, using a Raspberry Pi for media playback. Two video channels are converted into analog signals on UHF/VHF bands using HDMI-RF (radio frequency) modulators, transmitted through a set of coaxial cables to the external antenna ports of the television. A separate audio channel is converted and transmitted via radio signals using a low power FM transmitter.

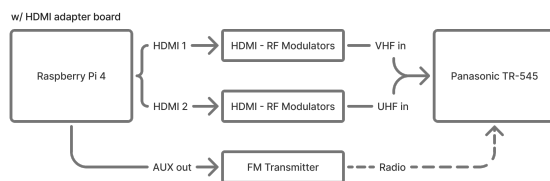


Figure 9: Installation diagram for video sculpture.

4.2 Visual Artifacts from Misaligned Channels

This setup intentionally mixes PAL analog signals with NTSC channels, resulting in partial alignment at their bandwidths' common multiples. Other than the aligned channels, testing reveals the fourth lower channel is always able to provide a stable display, but with a considerable amount of noise. Similar displays are also observed with other channels intermittently. The normally discarded frequency bands from the video signal's modulation process is restored and reclaimed as visual artifacts. By implementing the outdated technology in an incorrect way, the medium's normal functionality is disrupted, creating moments of failures.

The clear channel does not exist by itself, but among the distorted ones, reaffirming the project's core argument of normalcy, continuity, and the in-betweens.

5 Discussion

5.1 The Role of the Viewer

In the CRT video sculpture, viewers can switch channels, yet they receive no instructions, and some contain no signal—an ambiguity that invites hesitation. The ambient audio—a re-edited walkie-talkie reading of Borges' poem—appears on a specific FM frequency amid ordinary broadcasts. As they tune the dial, the voice emerges faint and fragmentary, surrounded by static, not as a dominant signal but as an intrusion into noise. In that fleeting crack within the everyday soundscape, night surfaces as a hidden temporal channel, briefly disrupting daylight's seamless continuity.

The web archive invites both disorientation and participation: visitors can submit new nighttime coordinates via a public form and community subreddits. When navigating these coordinates, there is no frame or endpoint; viewers must walk forward step by step, unsure when or whether daylight will return. The transition is abrupt, akin to waking.

The print album asks viewers to lift a transparent interface layer to reveal the screenshot beneath—a small physical act that stands in for removing the system's frame. Across components, the viewer shifts from consumer to critical witness, implicated in reproducing the glitch.

5.2 Feedback Loop as Governance

La Noche Cíclica does not end with documentation but enters a feedback loop. At the exhibition, a banner asks viewers whether to report nighttime coordinates to Google. This choice shifts the viewer from passive observer to participant, showing how users are drawn into platform optimization. GSV does not operate as a neutral service; it functions as a form of distributed algorithmic governance, subtly leveraging public action to maintain visual consistency. Reporting and replacement transform anomalies into infrastructural enforcement, where errors are ideologically suppressed. What appears as simple crowdsourcing is instead a mechanism disciplining what may be visible.

The project's origins highlight this process. Early coordinates came from Reddit communities such as *r/googlemapsshenanigans* and *r/geoguessr*, where anomalies were archived as curiosities. Once reported to Google, however, they risk deletion—such as the coordinate (40.6498945, -111.8349159, February 2020), later replaced with daylight. The tension between discovery and erasure transforms viewing into a form of governance, where visibility is not fixed, but negotiated and revised through infrastructural response. This project thus reframes feedback as a political gesture, rather than a simple technical refinement. Participatory correction becomes an invisible threshold of acceptability, continuously redrawn through user interaction.

5.3 Infrastructure and the American Night

While *La Noche Cíclica* appears to archive global anomalies, its contents are overwhelmingly drawn from the United States. This is not coincidental. As the primary testing ground and update hub for GSV, the U.S. receives more frequent and dense coverage than nearly any other region. These update cycles, though meant to maintain consistency, can produce stutters—temporal mismatches,



Figure 10: Displays on misaligned channels with different visual artifacts.

partial overlays, and unintentional ruptures. Paradoxically, it is in the places most surveilled and optimized where algorithmic inconsistencies most often appear.

This geographic concentration carries epistemological weight. The “night” captured in this project is not a universal condition, but a reflection of where Google’s infrastructural attention is most active. In contrast, vast regions of the Global South, rural zones, and conflict areas remain visually static—not due to stability, but because they fall outside the system’s rhythms of care. Here, absence is not a glitch but an infrastructural default. The American-centric anomalies in *La Noche Cíclica* thus underscore a larger truth: visibility itself is unevenly distributed, and error, too, follows infrastructural privilege.

5.4 Crowdsourcing and the Problem of Inclusion

Equally telling is what *La Noche Cíclica* excludes. Every coordinate in the archive is sourced from Google’s official panoramic imagery—not from crowdsourced “user-contributed photospheres” or blue-dot uploads. While many of these user submissions contain nighttime scenes—some eerie, others mundane—they exist outside the platform’s primary spatial-temporal logic. They are not stitched into the navigable continuum of Street View. Instead, they are peripheral entries, available only through direct interaction, marked as supplementary rather than canonical.

This distinction is crucial. User uploads are permitted precisely because they are excluded from the authoritative map. Their presence does not threaten infrastructural coherence, as they are siloed from the seamless flow of algorithmic vision. In contrast, the night coordinates chosen for this project are moments where the infrastructure itself falters—where Google unintentionally reveals what it tries to suppress.

By focusing solely on official imagery, *La Noche Cíclica* engages not with alternative or community-generated content, but with the system’s own blind spots. It frames night not as a product of user creativity or marginality, but as a systemic contradiction—one that arises within, not beyond, the apparatus of algorithmic control.

6 Conclusion

La Noche Cíclica invites audiences to reimagine visibility—not as something produced solely by data and daylight, but as something also shaped by gaps, glitches, and darkness. Infrastructures that once promised transparency are becoming increasingly opaque and totalizing; yet it is through their failures that a different kind of seeing becomes possible. These cracks—however fleeting—offer

moments to perceive otherwise. Through its interplay of visual research, archival disruption, and poetic speculation, this project reclaims night not merely as a temporal absence, but as a field of politics, perception, and potential.

By treating error as evidence and visibility as contested terrain, the project opens a methodological space for future inquiry into how algorithmic infrastructures can be read—not just functionally, but critically, fragmentally, and even poetically. In doing so, it suggests that artistic intervention may serve as both documentation and resistance within the evolving politics of machine vision.

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